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RESEARCH ARTICLE

Telemedicine Uptake among Older Adults in Small Towns: A Mixed-Methods Study

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ABSTRACT

Telemedicine consultations can improve access to healthcare for older residents of small towns, where the shortage of specialist physicians is particularly acute. However, uptake of such services among older people remains low, and the reasons are insufficiently understood. This mixed-methods study analyses the factors associated with telemedicine use among people aged 60 and over in four anonymised small towns. The quantitative phase comprised a telephone survey of 812 respondents; the qualitative phase consisted of 28 semi-structured interviews with patients and 9 interviews with healthcare workers. Over the past year, 19% of respondents had used telemedicine services. In a multivariable logistic model, the probability of use was associated with smartphone ownership, living with younger relatives and having chronic conditions requiring regular monitoring, whereas the association with age weakened after adjustment for these factors. Interviews revealed three key barriers: distrust of remote assessment of health status, complex booking and authentication procedures, and a fear of wasting the doctor's time without an in-person visit. Healthcare workers noted that telemedicine is most valuable for follow-up consultations and adjusting treatment. We conclude that simplified interfaces and support from nurses could substantially broaden telemedicine use among older adults.

Keywords: telemedicine; older adults; small towns; digital divide; mixed methods

Использование телемедицины пожилыми людьми в малых городах: исследование смешанными методами

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АННОТАЦИЯ

Телемедицинские консультации способны повысить доступность медицинской помощи для пожилых жителей малых городов, где нехватка врачей-специалистов особенно ощутима. Вместе с тем уровень использования таких услуг пожилыми людьми остаётся низким, а причины этого изучены недостаточно. В исследовании смешанными методами проанализированы факторы использования телемедицины людьми в возрасте 60 лет и старше в четырёх условных малых городах. Количественный этап включал телефонный опрос 812 респондентов, качественный — 28 полуструктурированных интервью с пациентами и 9 интервью с медицинскими работниками. За последний год телемедицинскими услугами воспользовались 19 % опрошенных. В многомерной логистической модели вероятность использования была связана с наличием смартфона, проживанием с родственниками младшего возраста и наличием хронических заболеваний, требующих регулярного наблюдения, тогда как связь с возрастом после учёта этих факторов ослабевала. Интервью выявили три ключевых барьера: недоверие к дистанционной оценке состояния здоровья, сложность процедур записи и авторизации, а также опасение «занять время врача» без очного визита. Медицинские работники отмечали, что наибольшую ценность телемедицина имеет для повторных консультаций и коррекции терапии. Сделан вывод о том, что упрощение интерфейсов и поддержка со стороны медицинских сестёр могут существенно расширить использование телемедицины пожилыми людьми.

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Ключевые слова: телемедицина; пожилые люди; малые города; цифровое неравенство; смешанные методы

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1 INTRODUCTION

Small towns face a persistent shortage of specialist physicians, and older residents often have to travel to regional centres for consultations. Telemedicine, in the form of video or telephone consultations with physicians, has been promoted as a way to bridge this gap, and the infrastructure for remote consultations expanded considerably after 2020.

Despite this, older adults, who stand to benefit most, use telemedicine less than other age groups. Explanations commonly focus on digital skills, but less attention has been paid to attitudes, household circumstances and the organisation of services. This study combines survey data and interviews to provide a fuller picture of telemedicine uptake among older residents of small towns.

2 MATERIALS AND METHODS

The quantitative phase was a computer-assisted telephone survey conducted in March–May 2024 in four towns with populations of 18,000–45,000 (coded T1–T4). A random sample of landline and mobile numbers yielded 812 completed interviews with residents aged 60 and over (response rate 37%). Telemedicine use in the past 12 months was the main outcome; covariates included demographics, health status, device ownership and household composition.

In the qualitative phase, 28 survey participants (14 users and 14 non-users) and 9 healthcare workers took part in semi-structured interviews. Transcripts were analysed thematically by two researchers independently, with discrepancies resolved by discussion.

3 RESULTS

Overall, 19% of respondents reported at least one telemedicine consultation in the past year. Uptake was highest among those with chronic conditions requiring regular follow-up (27%) and lowest among those living alone without a smartphone (6%) (Fig. 1).

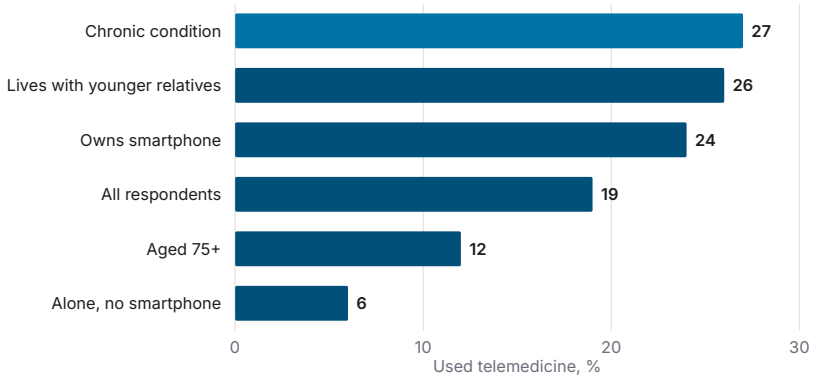


Fig. 1. Telemedicine use in the past 12 months by respondent group, %

Table 1 presents the adjusted odds ratios. Smartphone ownership more than tripled the odds of use, and living with younger relatives doubled them. The effect of age was attenuated and lost statistical significance after adjustment for device ownership and household composition.

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Table 1. Factors associated with telemedicine use (logistic regression, n = 812)

Factor	Adjusted OR	95% CI	p
Owns smartphone	3.4	2.1–5.5	<0.001
Lives with younger relatives	2.0	1.3–3.0	0.001
Chronic condition, regular monitoring	1.8	1.2–2.7	0.004
Age (per 5 years)	0.89	0.76–1.04	0.140
Female	1.1	0.8–1.6	0.520

4 DISCUSSION

Interviews helped interpret these associations. Younger relatives often acted as intermediaries, booking consultations and setting up devices. Non-users frequently expressed doubt that a physician could assess their condition without examination, and several described feeling that a remote consultation would be a burden on an already overstretched doctor.

Healthcare workers saw telemedicine as most useful for follow-up visits and medication adjustments, and suggested that nurses could help patients connect from the clinic or at home. These findings point to organisational rather than purely technological solutions.

5 CONCLUSION

Telemedicine uptake among older adults in small towns is limited less by age itself than by access to devices, household support and trust. Simplified booking procedures and nurse-facilitated consultations are promising ways to broaden its use.

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